RAMCO INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

Academic Year: 2019- 2020 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. CSE
Course Code & Title: CS8591 Computer Networks
Name of the Faculty members: Mrs.M.Swarna Sudha AP(CSE/CSE
Date & Time: 09.07.2019 & 50 minutes

Innovative practice: **Learning through Mind map**

Topic: Transmission media

Objectives:

O1: To familiarize the students with the process of Media Access
O2: Enhance the skill of visualizing and organizing the concept.

Justification for choosing the particular topic:
Transmission media topic have more key points, headings and sub heading which has to be remembered by students and then it would become overwhelming. But with Mind Map, makes the student to remember the sequence and prioritizing key topics.

Description:
A mind map is a diagram used to visually organize information. A mind map is hierarchical and shows relationships among pieces of the whole. It is often created around a single concept, drawn as an image in the center of a blank page, to which associated representations of ideas such as images, words and parts of words are added. From central idea radiate every idea that comes into our mind on that subject, dividing them into key ideas, or main brain branches. From main ideas/branches add r sub ideas, or child branches

Use of learning through argumentation

- Students will be able visually the concepts and organize information.
- Mind map helps students to relate the concepts.
- It can be used to classify ideas, and as an aid to study organizing information..
- It also allows students to remember key points and also how key points are connected directly to the central concept, and other ideas branch out from those central concepts.

Activity:
Teacher explains the particular concepts in class room within 30 minutes Based on the discussion and after clarifying the doubts by the students, teacher asked the students to draw a mind map related to the topic Each student draw a mind map based on their understanding level of the particular topic (Transmission media). This will help the students to recall the topic whichever discussed on the particular day, generate the new idea about the topic and they can easily answering the questions from this topic during exam by their own way.

Reflective report

Pre-implementation Reflection

Challenges:

- Some of the students represent very less key points in the mind map
- Presentation ideas of the mind map were not clearly represented.
- Some students felt difficult in representing the ideas in the mind map.

Steps taken for address the challenges

- Explained the uses of mind maps for topic and asked the students to represent all key points.
- Displayed the sample mind map in the class which includes all the key points.
- Once again explain the key concept which help the students to create the mind map easily.

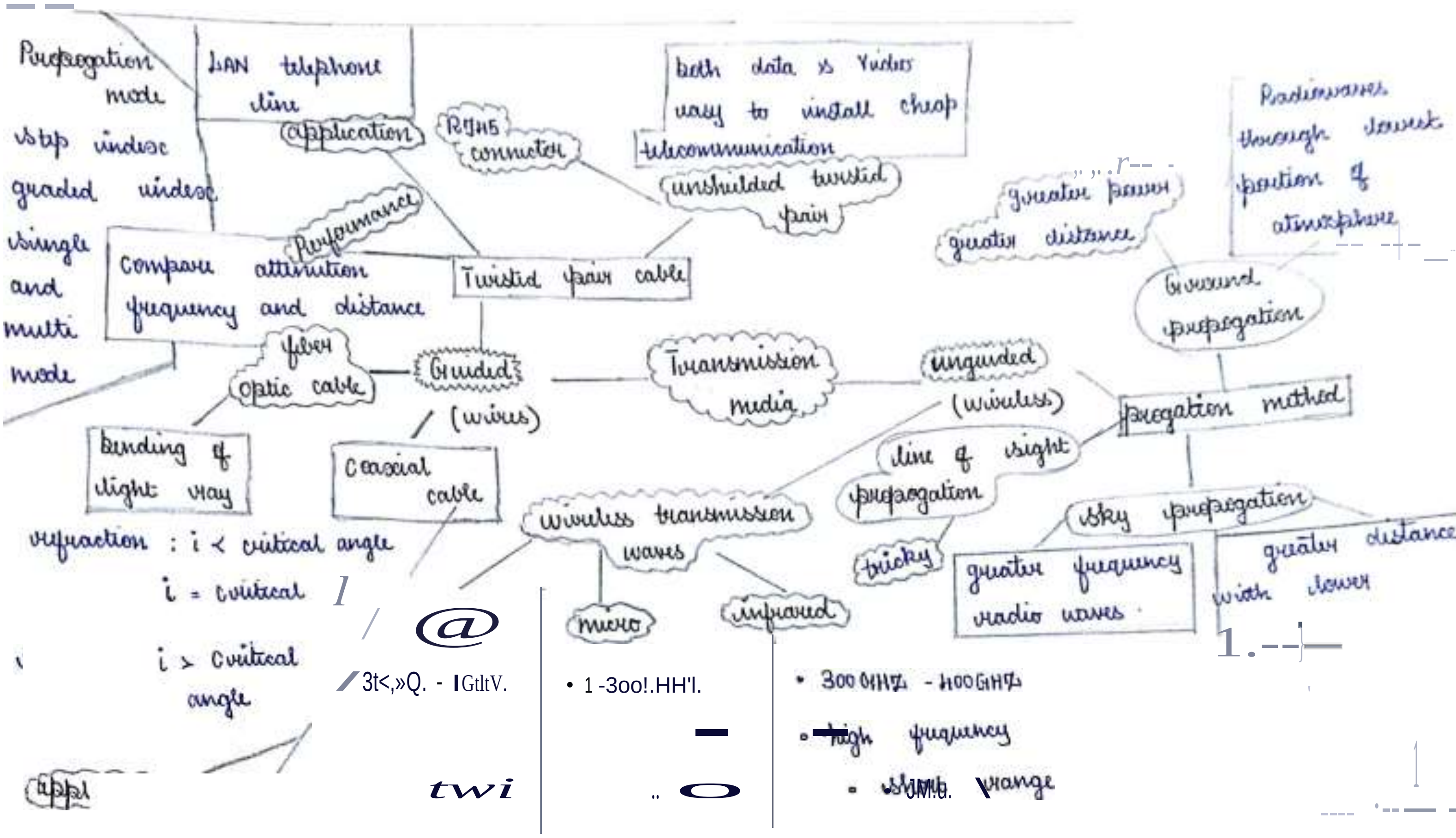
Post implementation reflection.

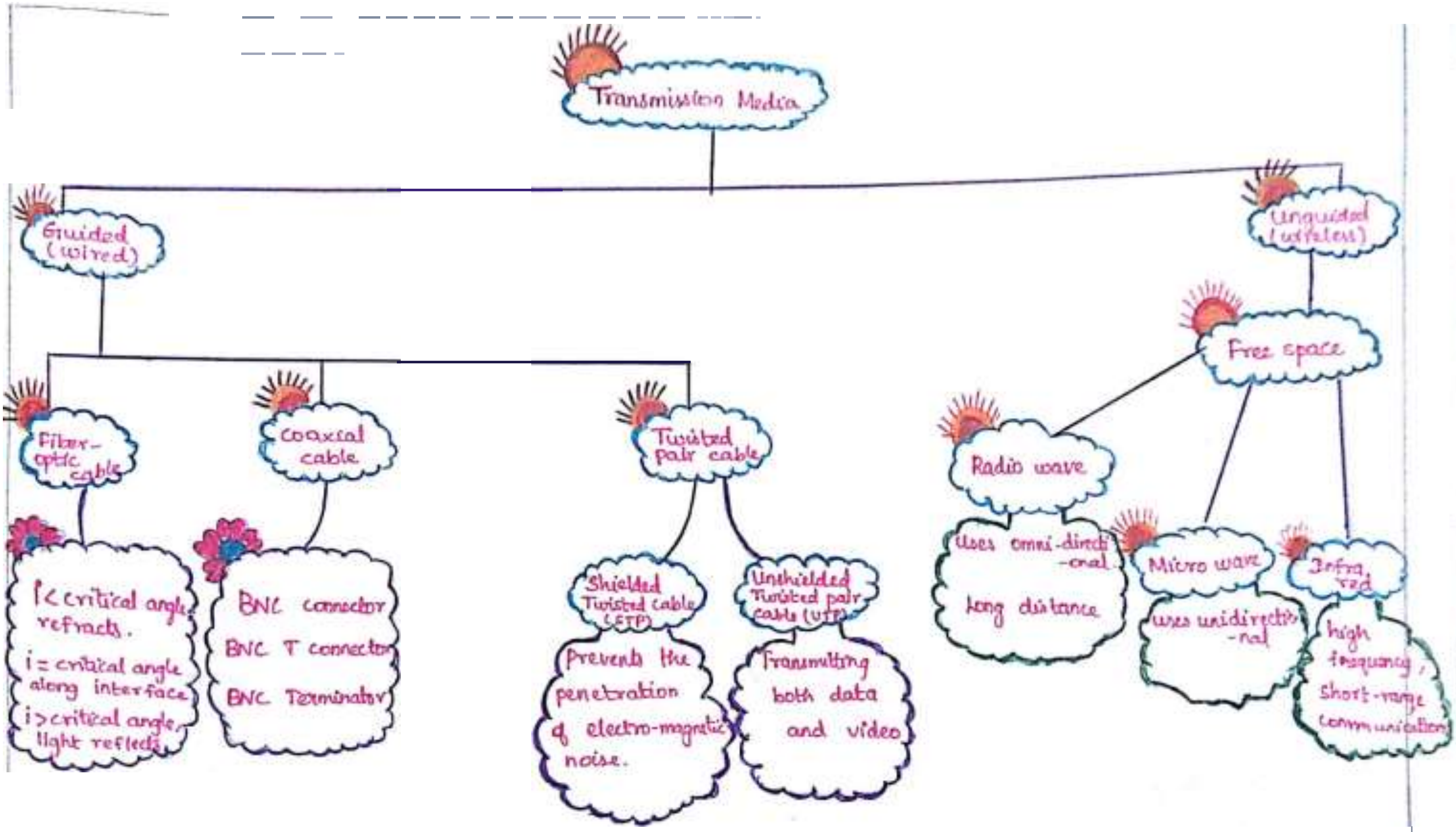
- Mind Maps are a great way for students to make notes guided and unguided wires on all of the information they receive.
- It help students to note down concepts of wired and wireless using key words, and then make connections between facts and ideas visually keeping all of transmission media thoughts together on one sheet
- It made key note making easier to students, as it reduces pages of notes into one single side of paper.
- Mind Map made students to remember the information more quickly.

This activity helps in attaining the following CO – PO mapping:

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
O1	3	1	2		2	1						1
O2	3			1								

Faculty In Charge **HOD**

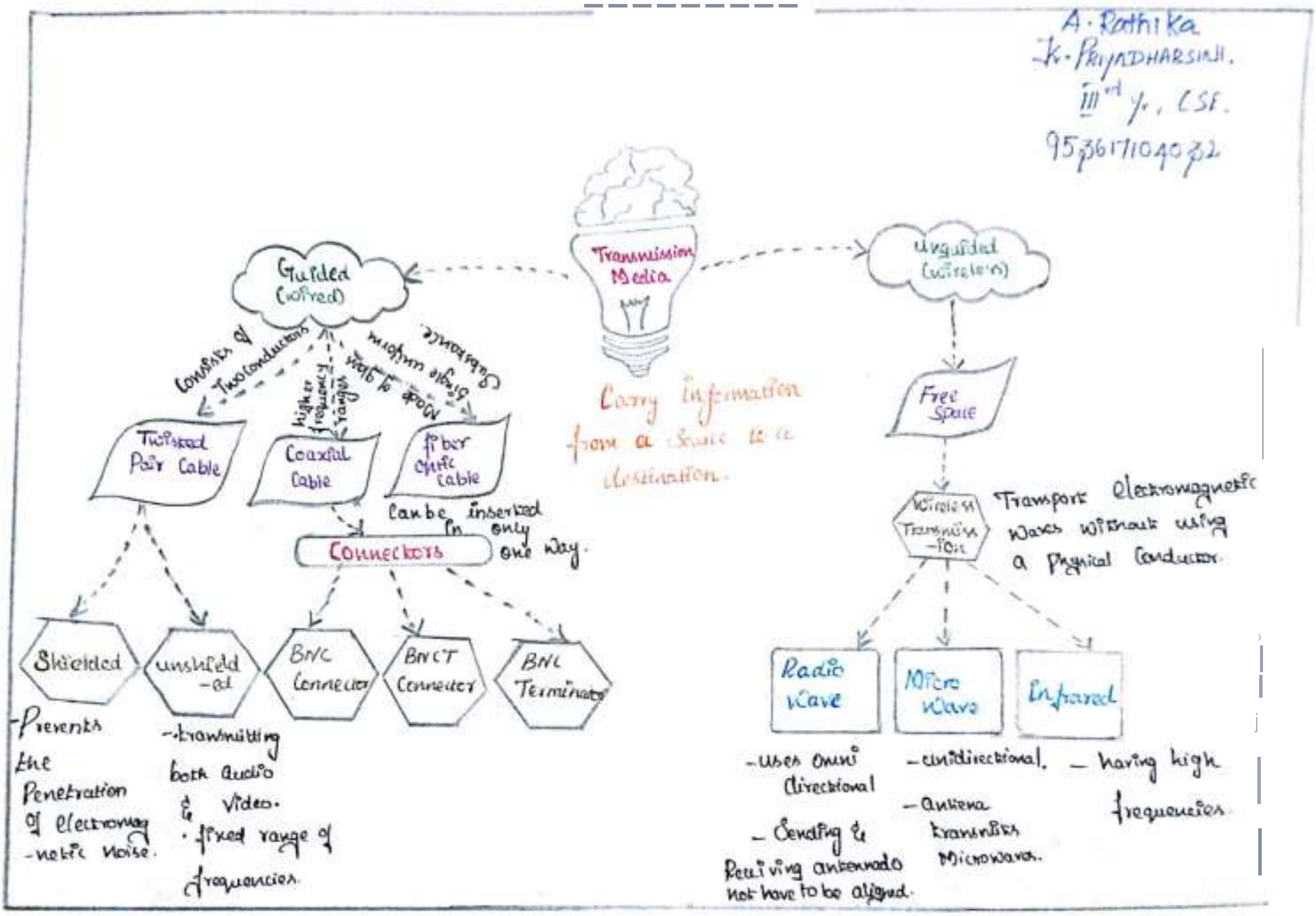

Propagation mode
step index
graded index
single and multi mode
LAN telephone line
application
RJ45 connector
both data & Video
easy to install cheap
telecommunication
unshielded twisted pair
Performance
compare attenuation frequency and distance
Twisted pair cable
fiber optic cable
Guided (wires)
Transmission media
unguided (wireless)
bending of light ray
Coaxial cable
refraction : i < critical angle
i = critical
i > critical angle
wireless transmission waves
micro
infrared
line of sight propagation
tricky
Propagation method
Ground propagation
greater power
greater distance
Radiowaves through lowest portion of atmosphere
sky propagation
greater frequency radio waves
greater distance with lower
300 GHz - 400 GHz
high frequency
short range


Transmission Media
Guided (wired)
Unguided (wireless)
Free space
Fiber-optic cable
Coaxial cable
Twisted pair cable
Radio wave
Micro wave
Infra red
i < critical angle refracts.
i = critical angle along interface
i > critical angle, light reflects
BNC connector
BNC T connector
BNC Terminator
Shielded Twisted cable (STP)
Unshielded Twisted pair cable (UTP)
Prevents the penetration of electro-magnetic noise.
Transmitting both data and video
uses omni-directi -onal
long distance
uses unidirectio -nal
high frequency, short-range communication


Transmission Media
Guided (wired)
Unguided (wireless)
Twisted pair cable
shielded twisted pair (STP)
unshielded twisted pair
expensive
bulkier
metal foil
RJ45 connector
cheap
telecommunication
greater power greater distance
Applications
telephone lines
performance
compare attenuation w/ frequency and distance
Fiber Optic Cable
propagation modes
multi mode
single mode
step index
graded index
connectors
subscriber channel
straight tip
MT-RJ
bending of light ray
refraction
reflection
critical angle
performance
o/p and o/p power.
Applications
backbone networks
hybrid networks
Coaxial cable
standards
connectors
BNC
BNC T
BNC termi-nator
RG-59
RG-58
RG-11
Application
analog telephone network
digital telephone network
performance
high attenuation
frequent use of repeaters
wireless transmission waves
*3KHz-1GHz
*1-300GHz
*omnidirectional
*long distance
*multicast communication
*unidirectional
*line of sight
*unicast communication
transports electromagnetic waves
wireless communication
Free space
Ground propagation
propagation methods
line of sight propagation
high frequency
Antenna to Antenna

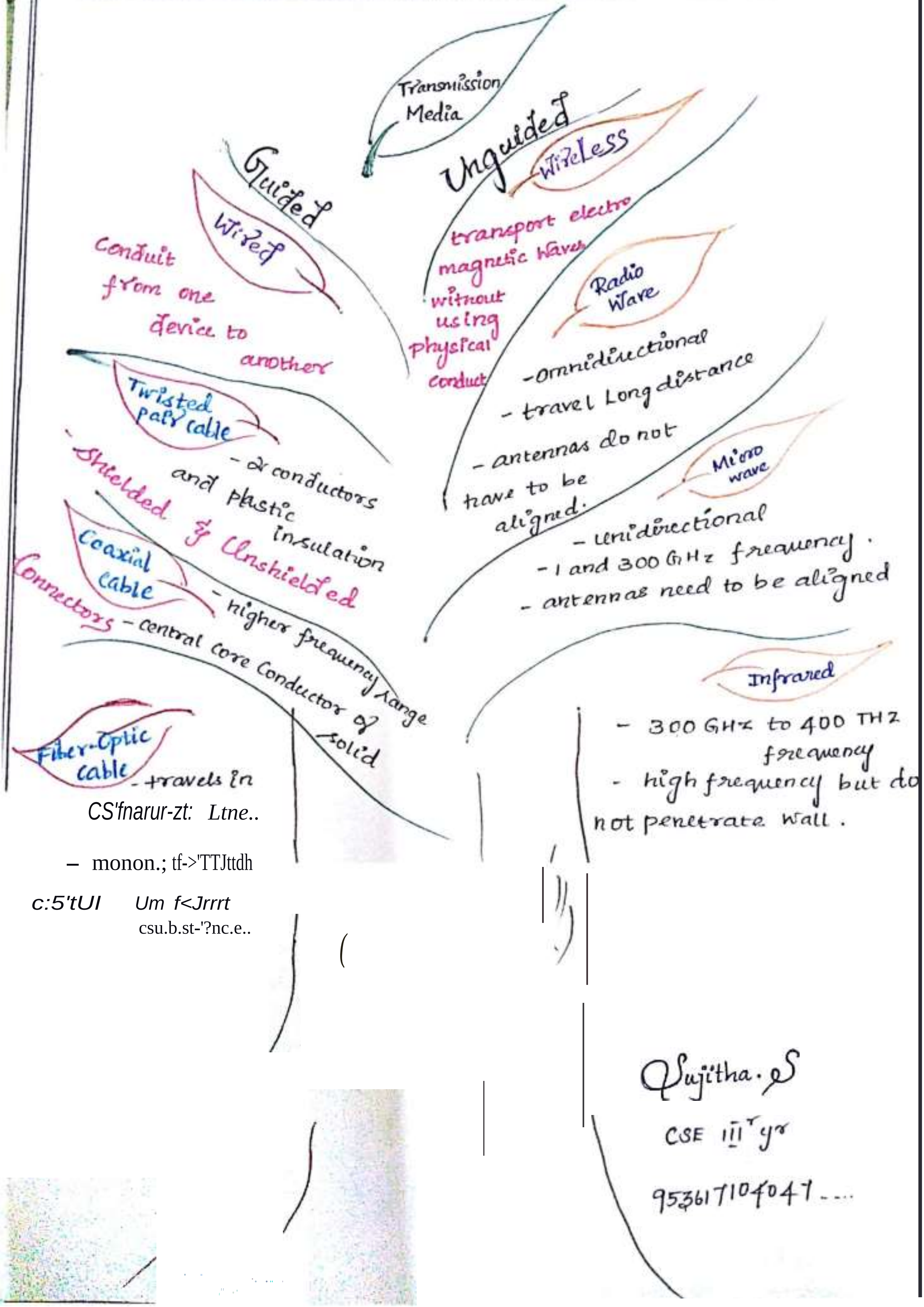
V. Varsha
16.7.19

* 3KHz - 1GHz	* 1-300 GHz	* 300 GHz to 400THz
* omnidirectional	* unidirectional	* high frequency
* long distance	* line of sight	* short range communication
* multicast communication	* unicast communication	

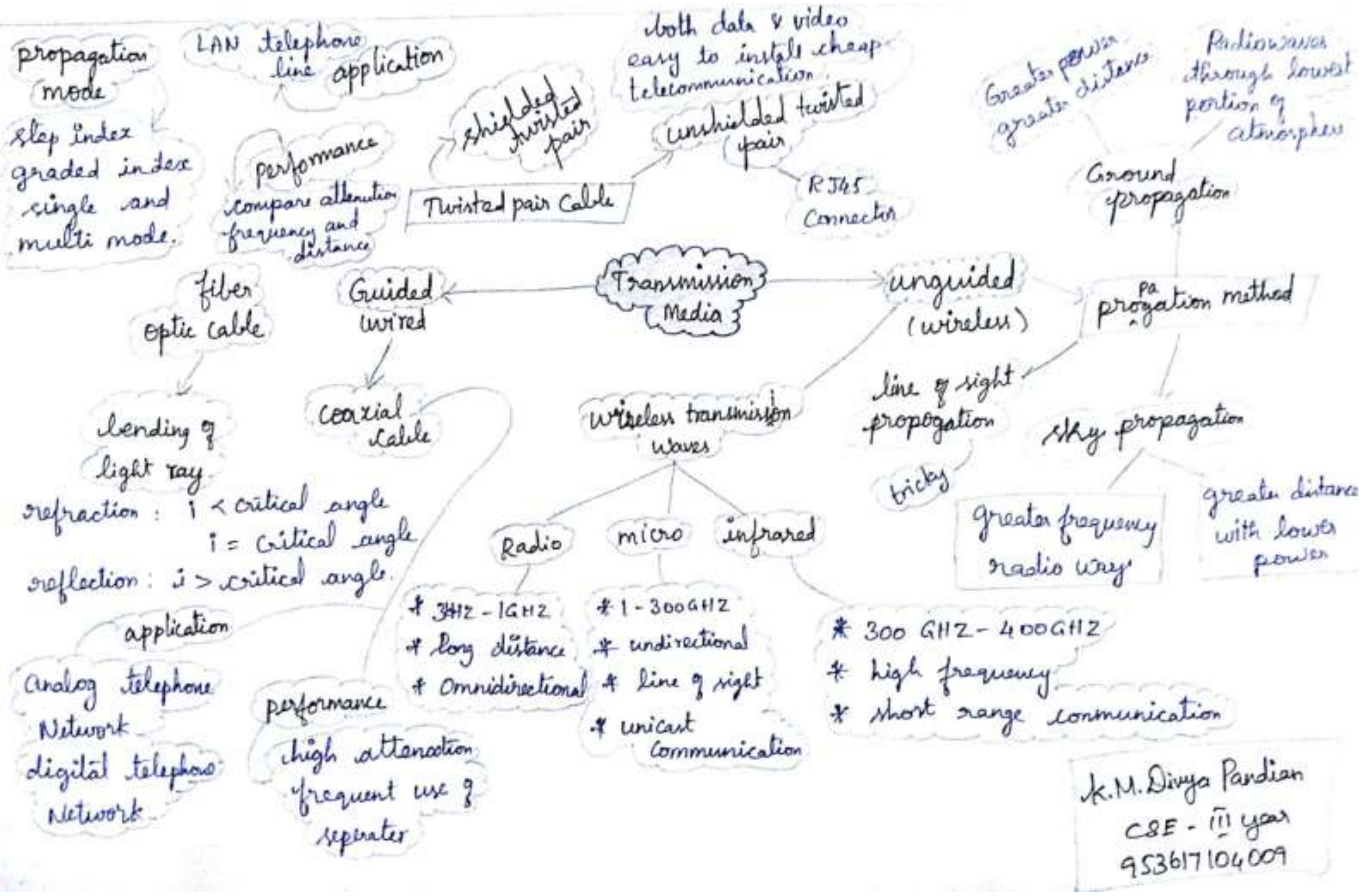


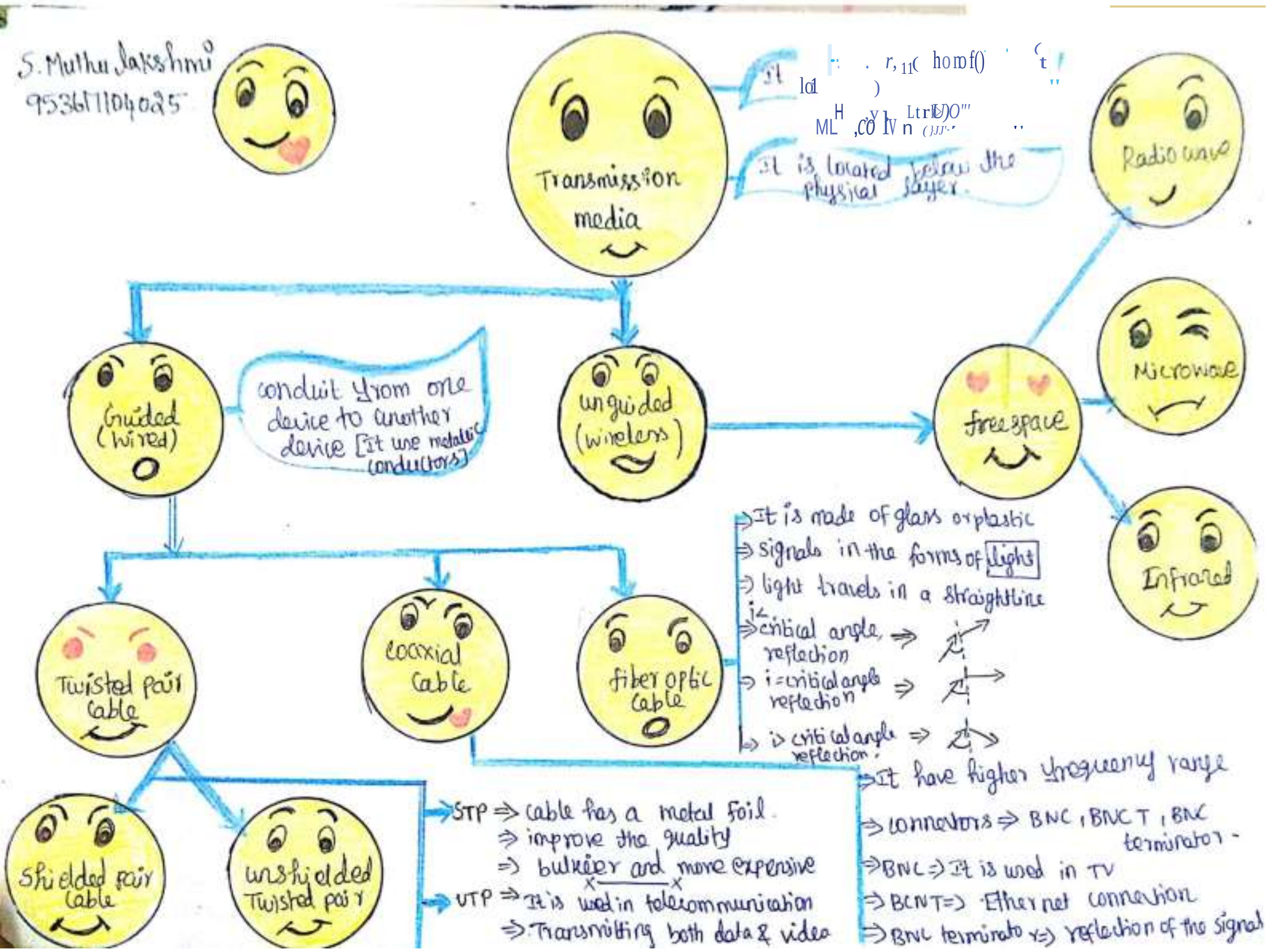
A. Rathika
K. PRIYADHARSINI.
III rd Yr., CSE.
9536171040 72
Transmission Media
Carry information from a Source to a destination.
Guided (Wired)
Unguided (Wireless)
Consists of Two conductors
higher frequency ranges
Made of glass
Single uniform Substance
Twisted Pair Cable
Coaxial Cable
Fiber optic cable
Can be inserted in only one way.
Connectors
Shielded
unshield -ed
BNC Connector
BNCT Connector
BNC Terminator
- Prevents the Penetration of electromag -netic noise.
- transmitting both audio & video.
· fixed range of frequencies.
Free Space
Wireless Transmiss -ion
Transport electromagnetic waves without using a Physical Conductor.
Radio wave
Micro wave
Infrared
- uses omni directional
- Sending & Receiving antenna do not have to be aligned.
- unidirectional.
- Antenna transmits Microwaves.
- having high frequencies.

COMPUTER NETWORKS

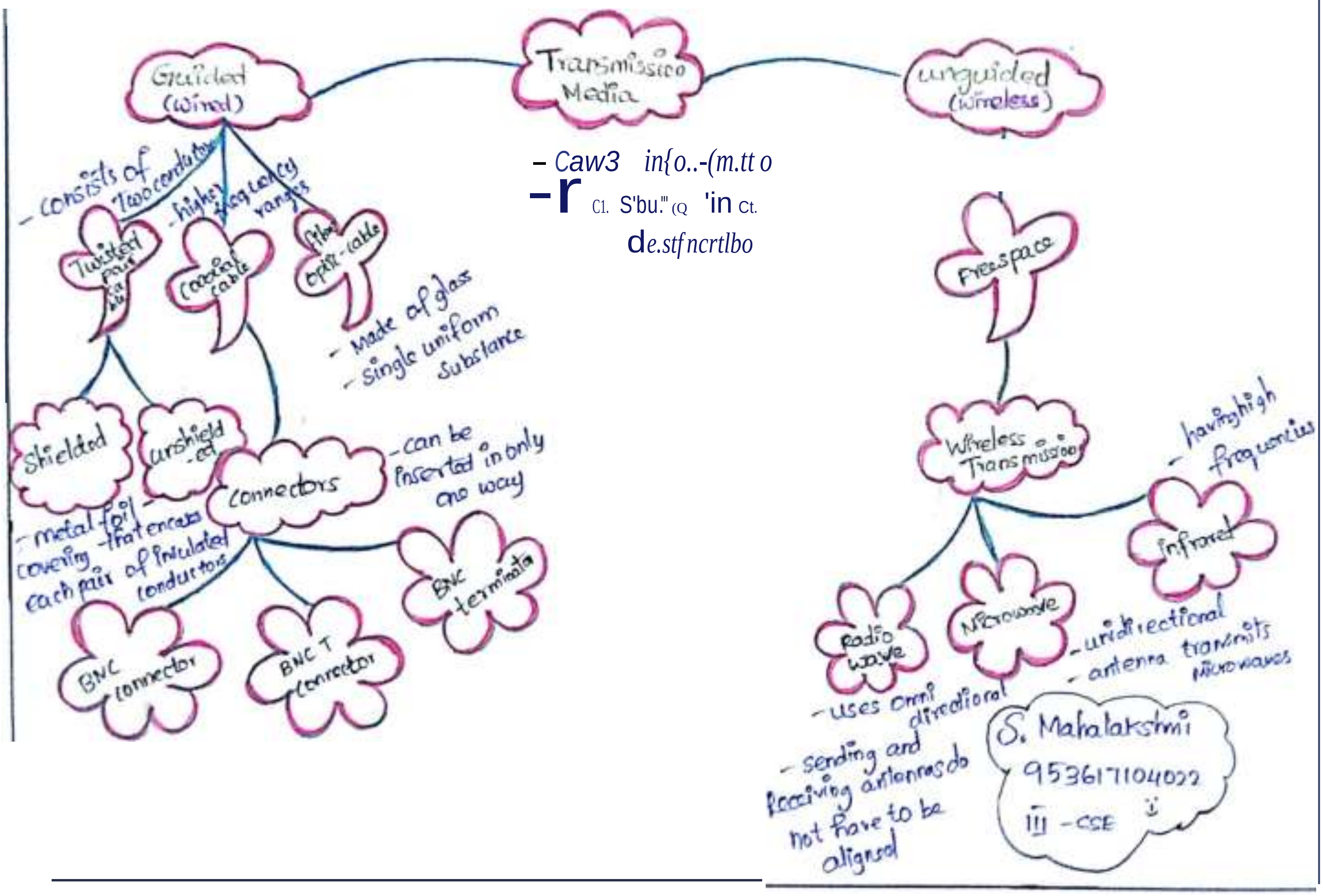


Sujitha. S
CSE III yr
9536171040 47...

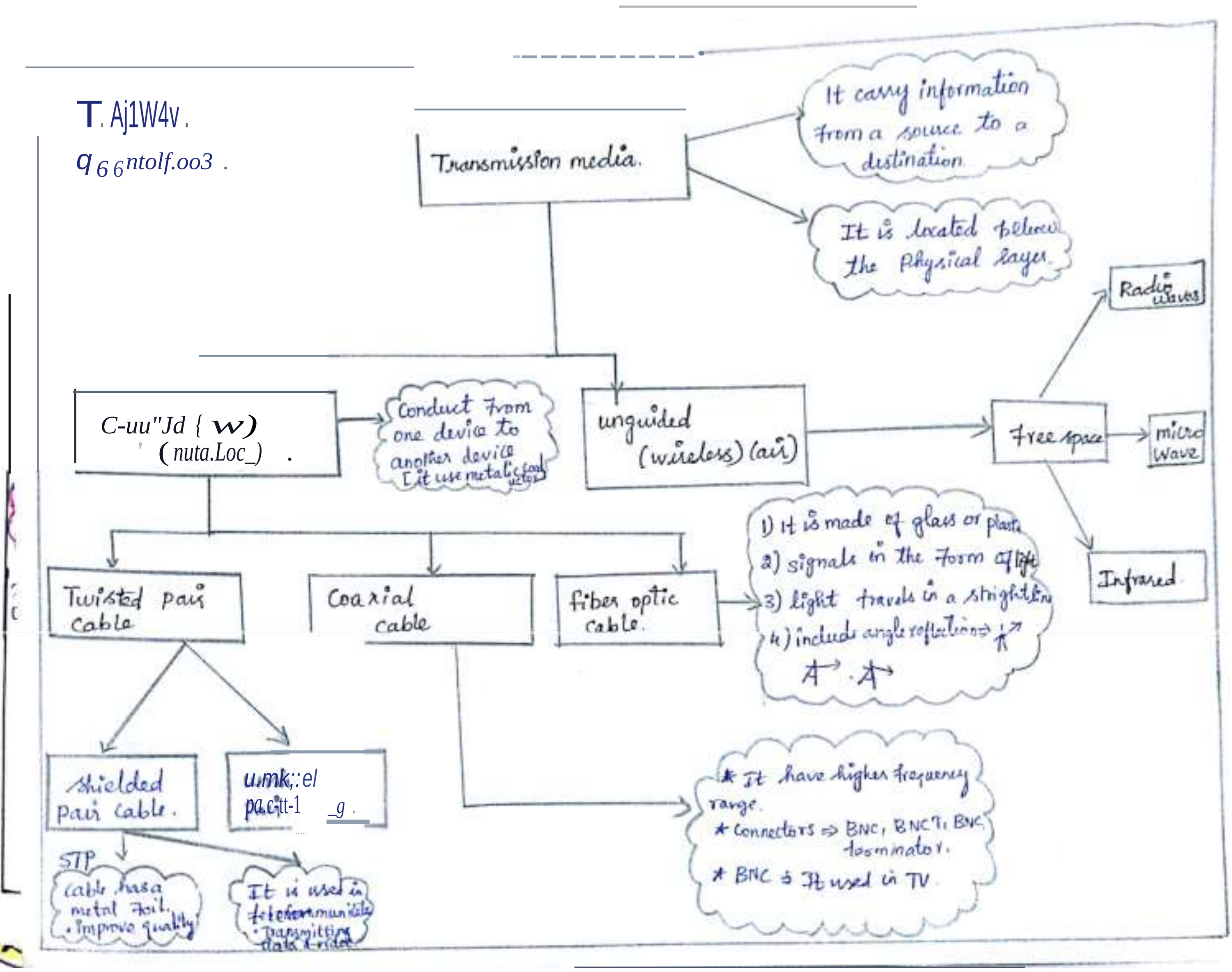

Transmission Media
Guided (wired)
unguided (wireless)
Twisted pair Cable
shielded twisted pair
unshielded twisted pair
RJ45 Connector
both data & video easy to install cheap telecommunication
LAN telephone line application
performance compare attenuation frequency and distance
fiber optic cable
propagation mode
step index graded index single and multi mode.
bending of light ray
refraction : i < critical angle
i = critical angle
reflection : i > critical angle
coaxial cable
application
analog telephone Network
digital telephone Network
performance
high attenuation frequent use of repeater
wireless transmission waves
Radio
micro
infrared
* 3HZ - 1GHZ
* long distance
* Omnidirectional
* 1 - 300GHZ
* unidirectional
* line of sight
* unicast Communication
* 300 GHZ - 400GHZ
* high frequency
* short range communication
propagation method
Ground propagation
Greater power greater distance
Radiowaves through lowest portion of atmosphere
line of sight propagation
tricky
sky propagation
greater frequency radio wave
greater distance with lower power
K.M. Divya Pandian
CSE - III year
953617104009

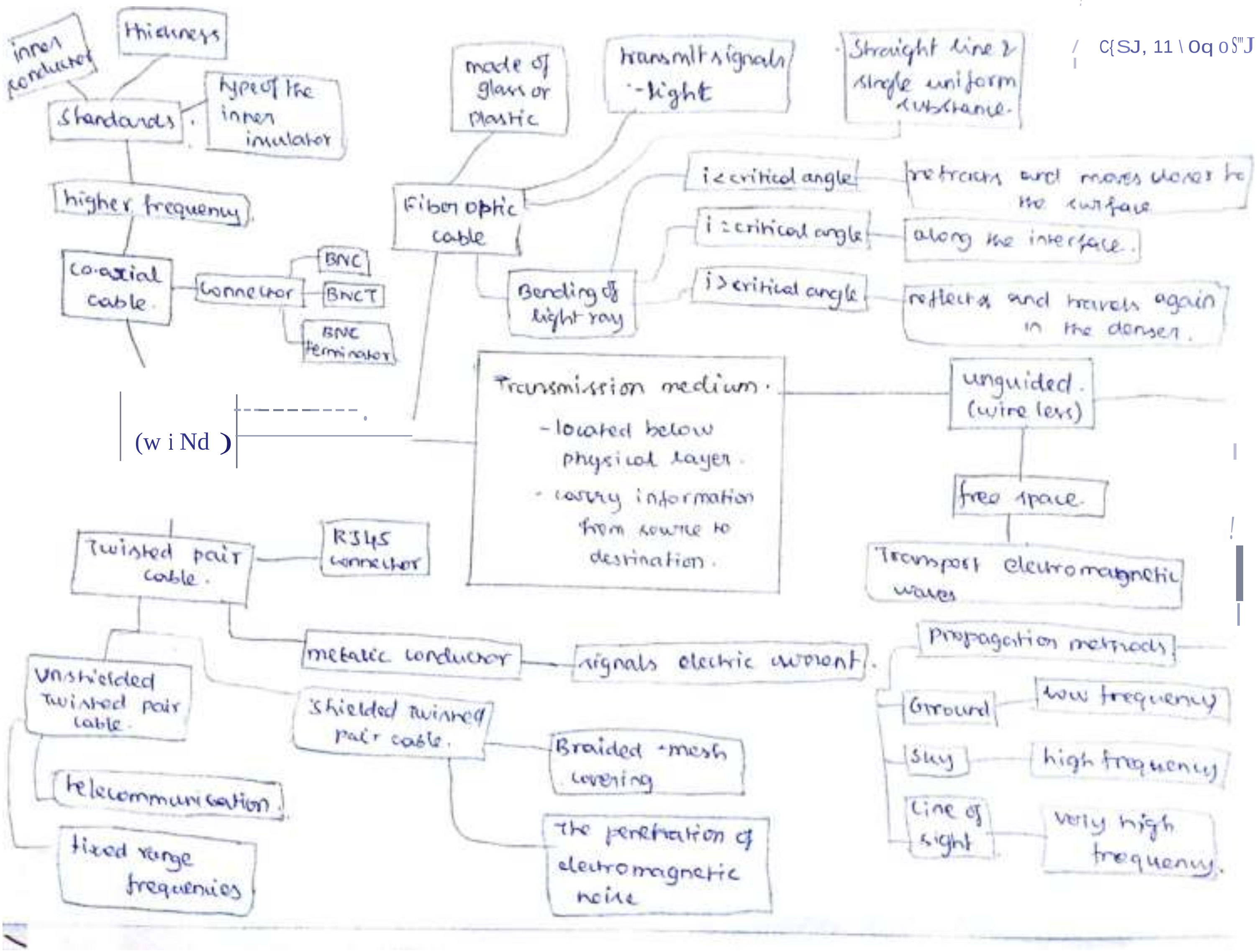


S. Muthu lakshmi
95361104025
Transmission media
It is located below the physical layer.
Guided (wired)
conduit from one device to another device [It use metallic conductors]
unguided (wireless)
free space
Radio wave
Microwave
Infrared
Twisted pair cable
coaxial cable
fiber optic cable
⇒ It is made of glass or plastic
⇒ signals in the forms of light
⇒ light travels in a straight line
⇒ i < critical angle ⇒ reflection
⇒ i = critical angle reflection ⇒
⇒ i > critical angle ⇒ reflection
⇒ It have higher frequency range
⇒ connectors ⇒ BNC, BNC T, BNC terminator
⇒ BNC ⇒ It is used in TV
⇒ BCNT ⇒ Ethernet connection
⇒ BNC terminator ⇒ reflection of the signal
Shielded pair cable
unshielded Twisted pair
STP ⇒ cable has a metal foil
⇒ improve the quality
⇒ bulkier and more expensive
UTP ⇒ It is used in telecommunication
⇒ Transmitting both data & video



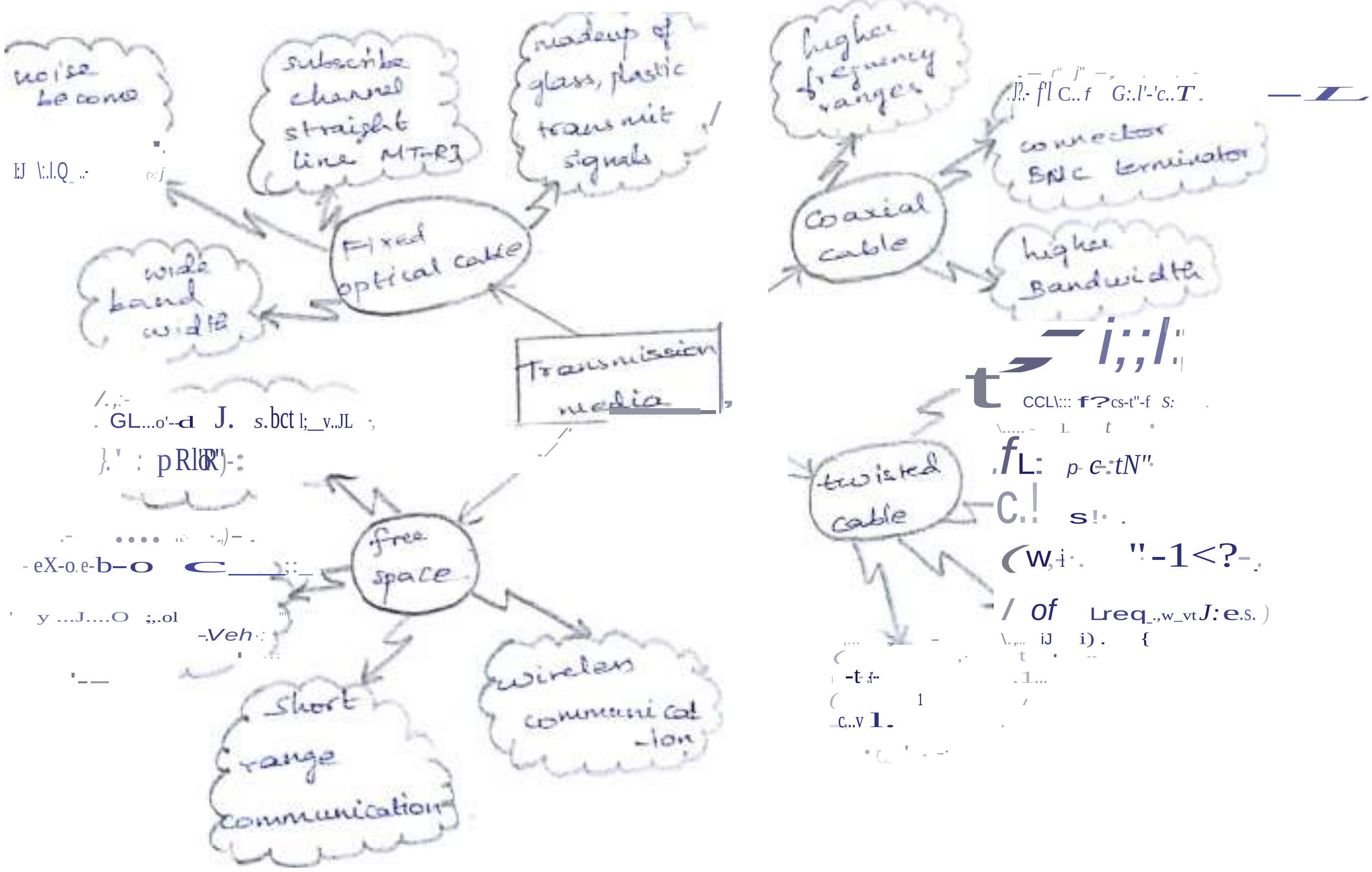
Transmission Media
Guided (Wired)
Unguided (Wireless)
- consists of Two conductors
- higher frequency ranges
Twisted pair cable
Coaxial cable
Fiber optic cable
- Made of glass
- Single uniform substance
Shielded
unshield -ed
- metal foil covering that encases each pair of insulated conductors
Connectors
- can be inserted in only one way
BNC connector
BNC T connector
BNC terminator
Free space
Wireless Transmission
Radio wave
Microwave
Infrared
- having high frequencies
- unidirectional antenna transmits microwaves
- uses omni directional
- sending and receiving antennas do not have to be aligned
S. Mahalakshmi
953617104022
III - CSE


Transmission media.
It carry information from a source to a destination
It is located below the Physical layer.
Conduct from one device to another device
unguided (wireless) (air)
Free space
Radio waves
micro Wave
Infrared.
Twisted pair cable
Coaxial cable
fiber optic cable.
1) It is made of glass or plastic
2) signals in the form of light
3) light travels in a straight line
4) include angle reflection
Shielded Pair cable.
STP
cable has a metal foil.
• Improve quality
It is used in telecommunication
• Transmitting data & video
* It have higher frequency range.
* Connectors ⇒ BNC, BNC T, BNC terminator.
* BNC ⇒ It used in TV



inner conductor
thickness
type of the inner insulator
Standards
higher frequency
co-axial cable
connector
BNC
BNCT
BNC terminator
(w i Nd)
made of glass or plastic
transmit signals - light
Straight line & single uniform substance.
Fiber optic cable
i < critical angle
refracts and moves closer to the surface
i = critical angle
along the interface.
i > critical angle
reflects and travels again in the denser.
Bending of light ray
Transmission medium.
- located below physical layer.
- carry information from source to destination.
unguided. (wire less)
free space.
Transport electromagnetic waves
Propagation methods
Ground
low frequency
Sky
high frequency
Line of sight
very high frequency.
Twisted pair cable.
RJ45 connector
metallic conductor
signals electric current.
Unshielded twisted pair cable.
telecommunication
fixed range frequencies
Shielded twisted pair cable.
Braided mesh covering
The penetration of electromagnetic noise

S. NARMADHA 953617104027




Transmission media
Guided (wired)
Unguided (wireless)
Twisted pair cable
Shielded twisted pair (STP)
expensive bulkier metal foil
Unshielded twisted pair (UTP)
RJ45 connector
both data & video easy to install cheap tele communication
telephone system
suitable for voice & data transmission upto 10 mbps, data transmission upto 20 mbps, data transmission upto 100 mbps
category
Application
LAN telephone wires
performance
compare attenuation vs frequency and distance
Fiber optic cable
single mode
multi mode
step index
graded index
propagation modes
connectors
subscriber channel straight tip MT-RJ
bending of light ray
refraction i < critical angle
i = critical angle
reflection i > critical angle
Applications
backbone networks
hybrid networks
Coaxial cable
standards
RG-59
RG-58
RG-11
connectors
BNC
BNC T
BNC terminator
Application
analog telephone network
digital telephone network
wireless transmission waves
radio waves
micro wave
infra red
3 KHz - 1 GHz
omnidirectional
long distance
multicast communication
1-300 GHz
unidirectional
line of sight
unicast communication
300 GHz to 400 THz
high frequency
short range communication
transport electromagnetic wireless communication free space
propagation methods
ground propagation
through lowest portion of atmosphere
greater power greater distance
sky propagation
higher frequency radio waves
greater distance with lower power
line of sight propagation
high frequency antenna to antenna

2. unshielded

Performance
compare attenuation
versus distance and frequency
measured in db/km
pass wide range of frequency

Application
used in
telephone
lines.

Propogation mode
single mode
multimode

Twisted pair
consist of two conductors
twisted together

connectors
sc connector
ST connector
MT-RJ connector

fibre optic cable
transmit signal in
form of light

Performance
compare input power to
output power
high bit error rates
noise become problem

coaxial
cable
carries
signal of
high
frequency

Guided media
Provide a conduit
from one device to
another.

Transmission medium
carries information
from source to
destination

unguided media
transport em
waves without
physical conductor

Propogation method
Ground propagation
sky propagation
line-of-sight
propagation

connector
BNC

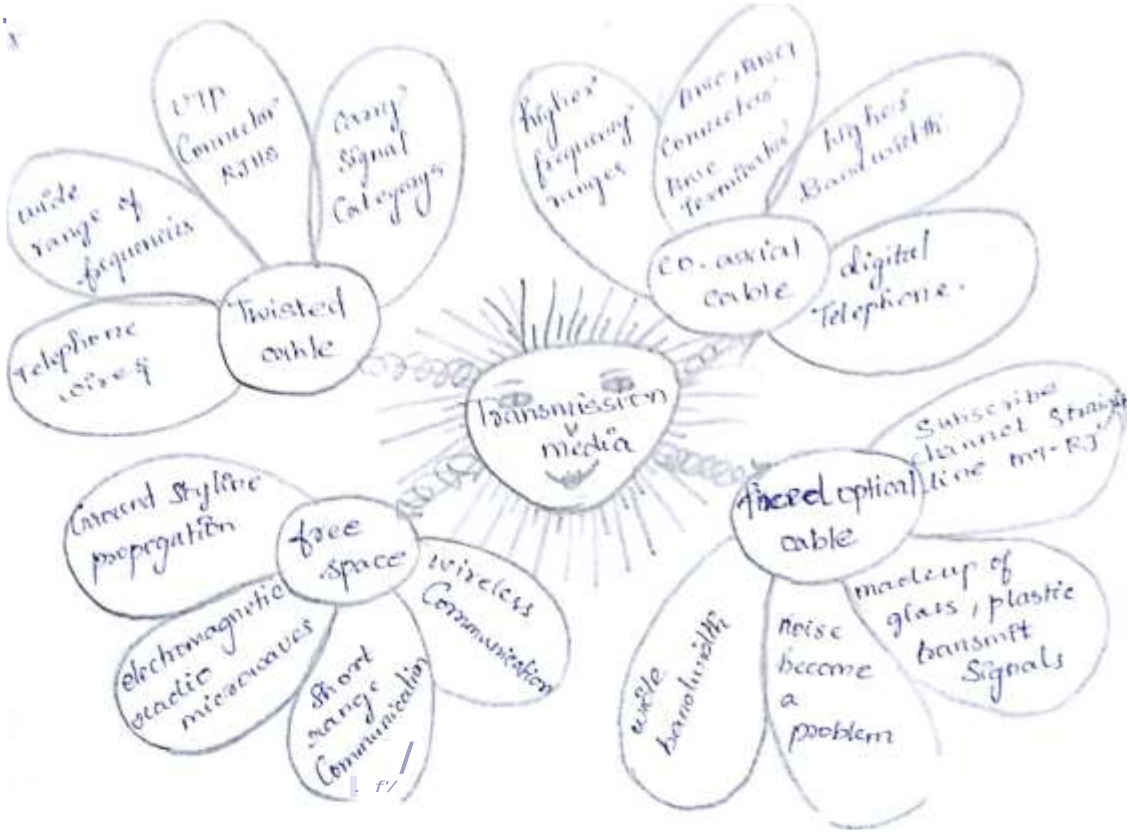
Performance
Attenuation is
much higher
higher bandwidth

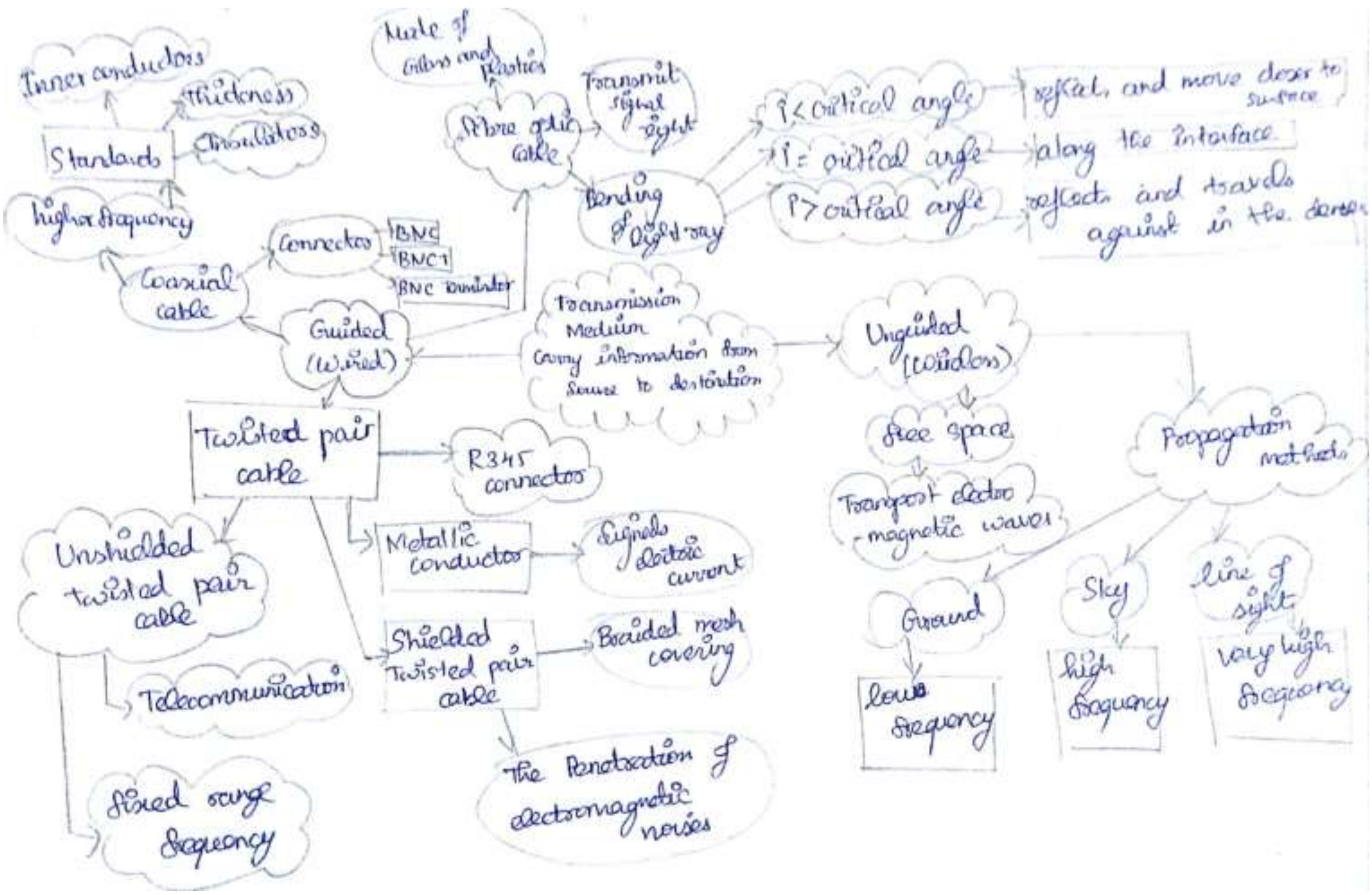
Application
widely used
in analog
telephone
networks.

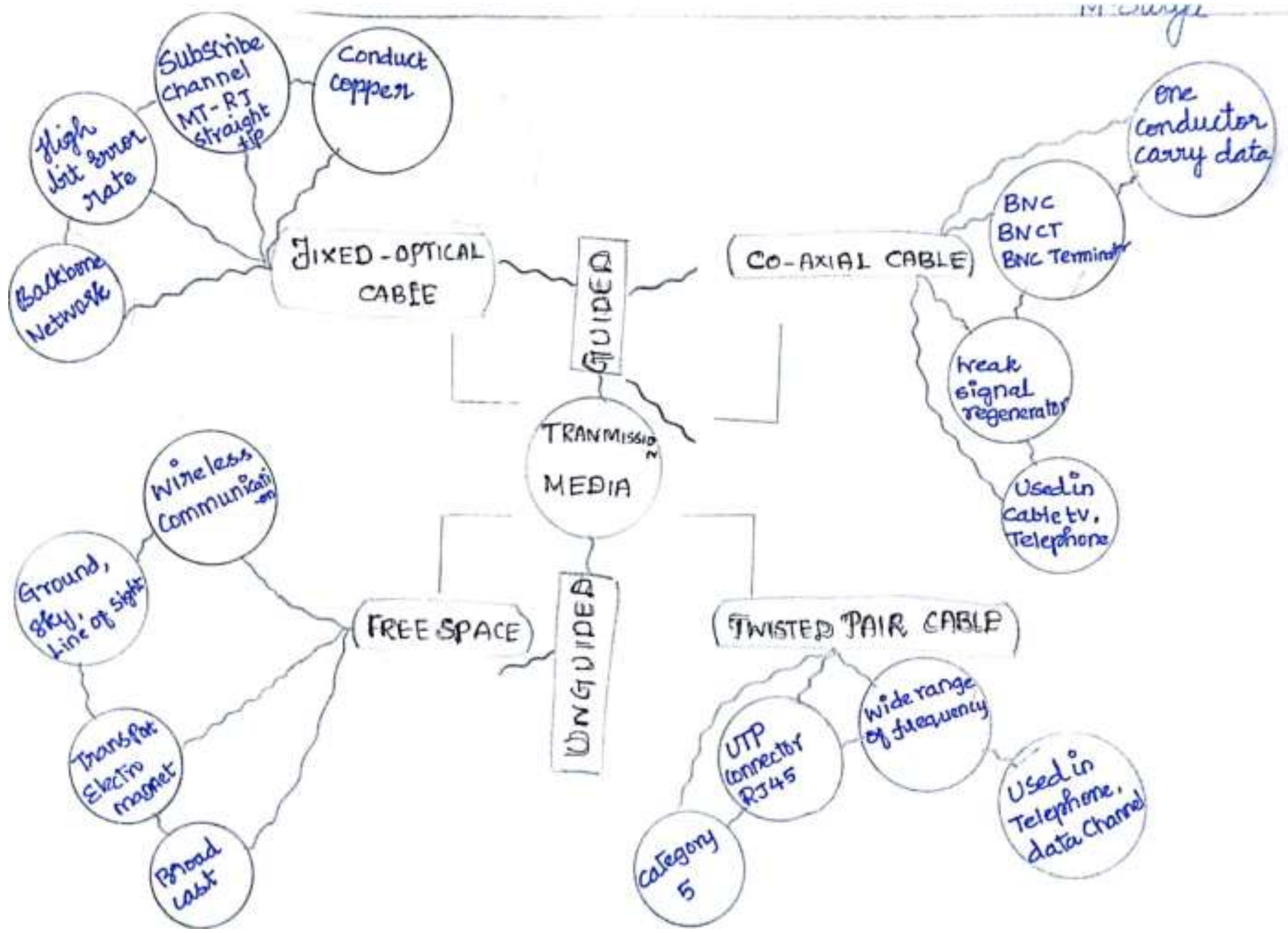
Radio waves
Omnidirectional
can travel long
distance
can receive signal
inside buildings
multicast
communication

Microwaves
unicast
communication
High frequency
cannot penetrate
Microwave band -
299 GHZ
frequency range
1-300 GHz

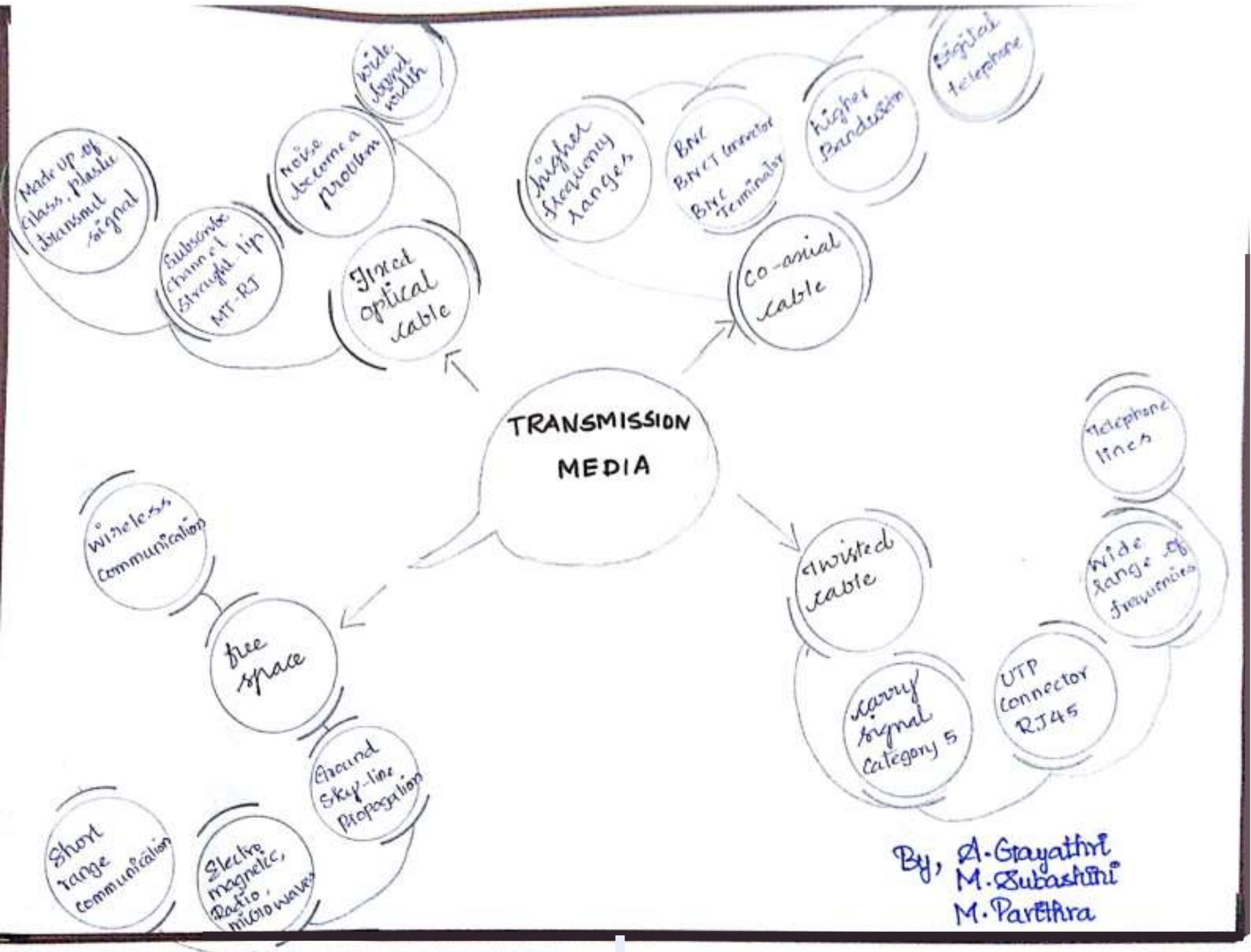
Infrared waves
frequency range
300 GHz - 400 TH
short range
communication
uses line-of-sig
propagation


Transmission media
Twisted cable
UTP Connector RJ45
Carry Signal Category
wide range of frequencies
Telephone wires
Co-axial cable
higher frequency ranges
higher Bandwidth
digital Telephone.
fibre optical cable
wide bandwidth
noise become a problem
madeup of glass, plastic transmit Signals
free space
Ground Styline propagation
wireless Communication
Short range Communication
electromagnetic radio microwaves

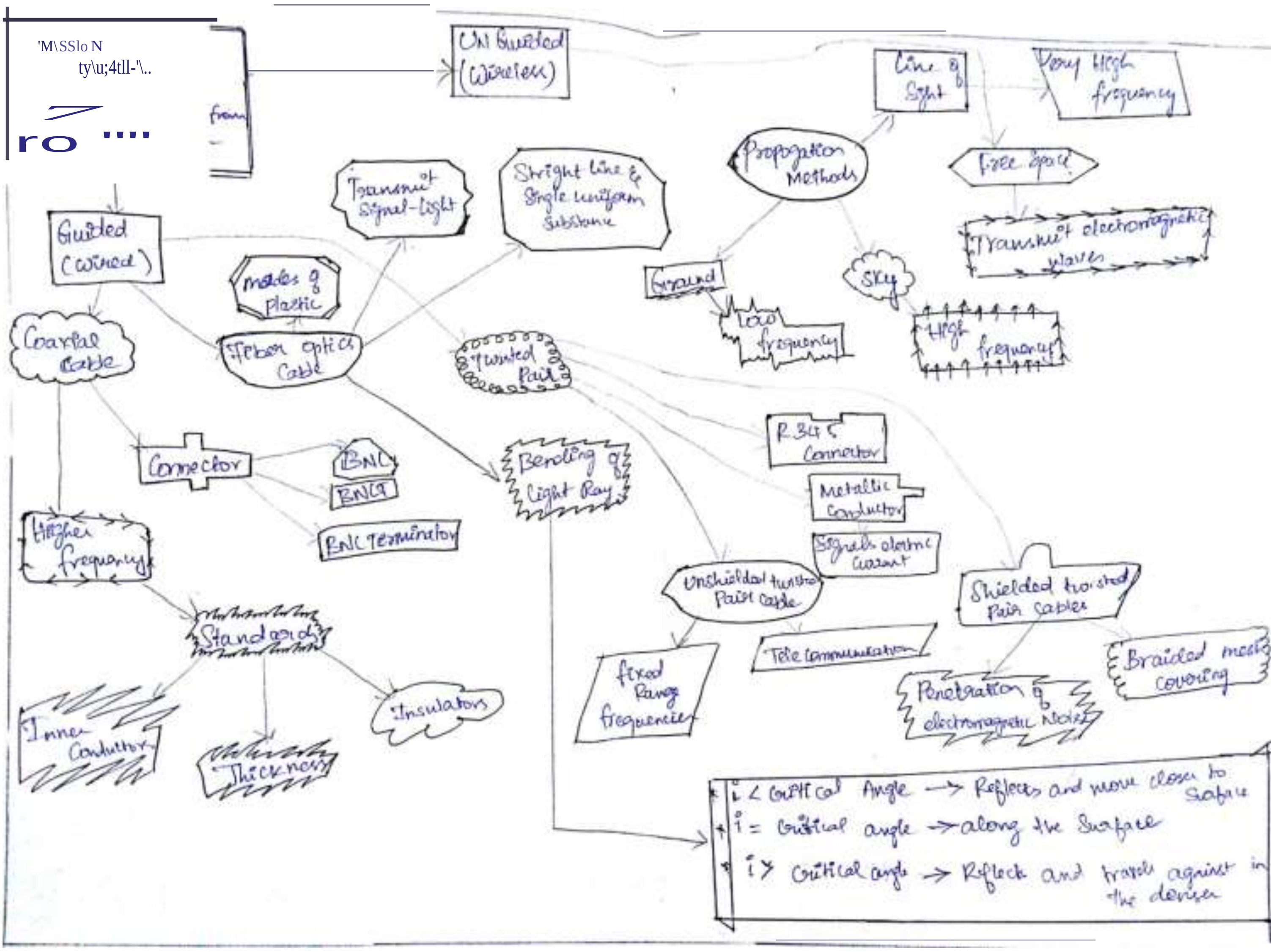

Inner conductors
thickness
Standards
Insulators
higher frequency
Coaxial cable
Connectors
BNC
BNC T
BNC terminator
Made of Glass and Plastics
Fibre optic cable
Transmit signal light
Bending of light ray
i < critical angle
refract and move closer to surface
i = critical angle
along the interface
i > critical angle
reflect and towards against in the denser
Transmission Medium
Carry information from Source to destination
Guided (Wired)
Unguided (Wireless)
Twisted pair cable
RJ45 connector
Metallic conductor
Signals electric current
Shielded Twisted pair cable
Braided mesh covering
The Protection of electromagnetic noises
Unshielded twisted pair cable
Telecommunication
fixed range frequency
free space
Transport electro-magnetic waves
Propagation methods
Ground
Sky
line of sight
low frequency
high frequency
very high frequency



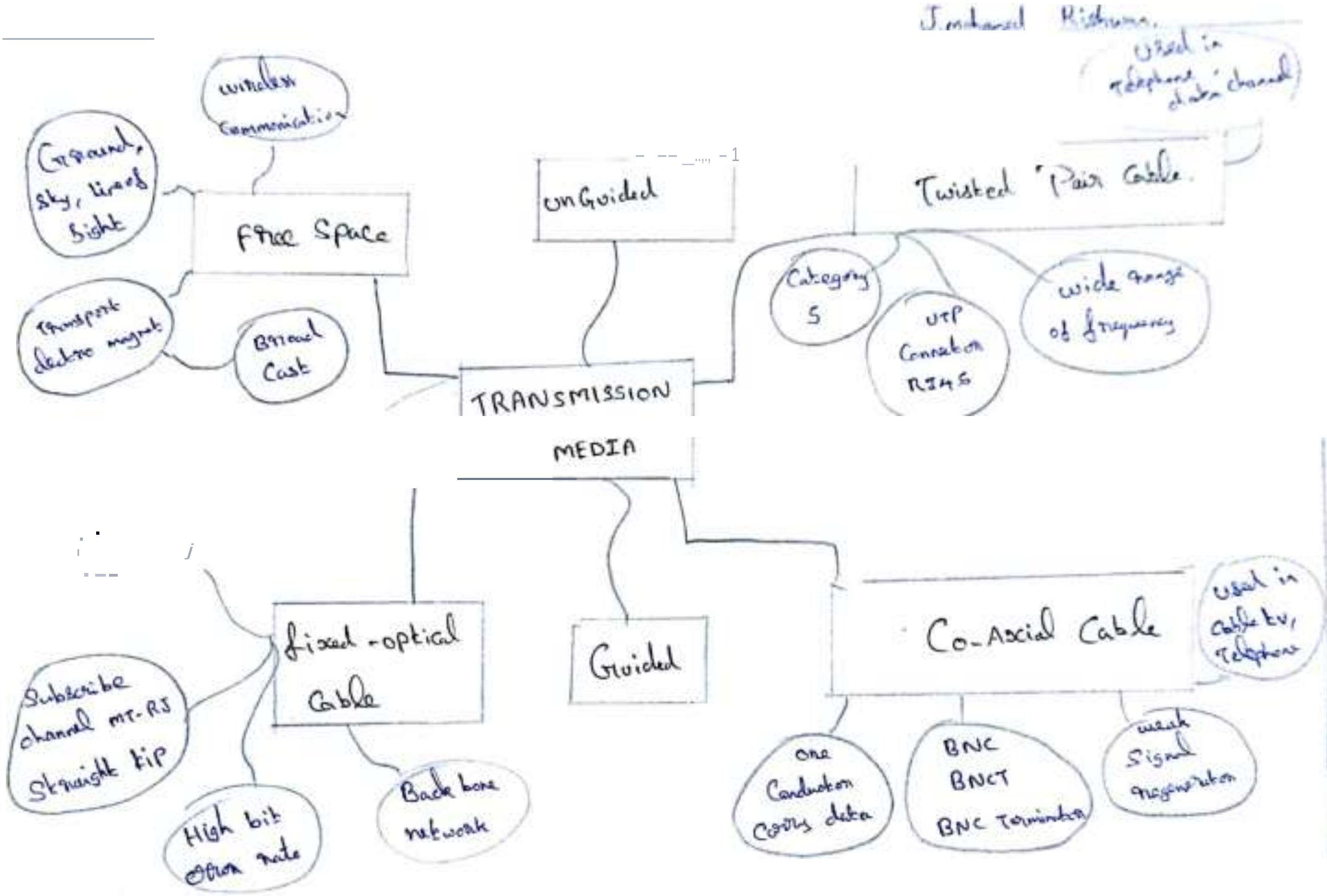
TRANMISSION MEDIA
GUIDED
UNGUIDED
FIXED-OPTICAL CABLE
Conduct copper
Subscribe channel MT-RJ straight tip
High bit error rate
Backbone Network
CO-AXIAL CABLE
One conductor carry data
BNC BNCT BNC Terminator
Weak signal regenerator
Used in Cable tv, Telephone
FREE SPACE
Wireless Communication
Ground, sky, Line of sight
Transport Electro magnet
Broad cast
TWISTED PAIR CABLE
UTP connector RJ45
Wide range of frequency
Used in Telephone, data channel
category 5



TRANSMISSION MEDIA
Fixed optical cable
Made up of Glass, plastic transmit signal
Subscribe channel Straight tip MT-RJ
Noise become a problem
wide band width
Co-axial cable
higher frequency ranges
BNC BNCT connector BNC Terminator
higher Bandwidth
Digital telephone
Twisted cable
Telephone lines
wide range of frequencies
UTP connector RJ45
carry signal category 5
free space
wireless communication
Ground Sky-line Propagation
Electro magnetic, Radio, micro waves
Short range communication
By, A.Gayathri M.Subashini M.Pavithra



Un Guided (Wireless)
Guided (Wired)
Line of Sight
Very High frequency
Propagation Methods
Free space
Transmit electromagnetic waves
Ground
Sky
Low frequency
High frequency
Transmit Signal-Light
Stright Line & Single uniform Substance
Modes of Plastic
Coaxial Cable
Fiber optics Cable
Twisted Pair
Connector
BNC
BNCT
BNC Terminator
Bending of Light Ray
RJ45 Connector
Metallic Conductor
Signals electric Current
Higher frequency
Standards
Inner Conductor
Thickness
Insulators
Unshielded twisted Pair cable
Shielded twisted Pair cables
Tele communication
fixed Range frequencies
Penetration of electromagnetic Noise
Braided mesh covering
i < Critical Angle → Reflects and move closer to Surface
i = Critical angle → along the Surface
i > Critical angle → Reflects and travels against in the denser


TRANSMISSION MEDIA
unGuided
Free Space
wireless Communication
Ground, sky, line of sight
Transport electro magnet
Broad Cast
Twisted Pair Cable
Used in telephone data channel
Category 5
UTP Connector RJ45
wide range of frequency
Guided
fixed-optical Cable
Subscribe channel MT-RJ Straight tip
High bit error rate
Back bone network
Co-Axial Cable
used in cable tv, Telephone
one Conductor carry data
BNC
BNCT
BNC Terminator
weak Signal regeneration